

Parser Tables & Grammar Analysis

1. FIRST & FOLLOW Sets

Non-Terminal	FIRST Set	FOLLOW Set
ActualParams	{ (, epsilon }	{), *, +, ,, -, /, :, <, ≤, ◇, =, >, ≥,], and, div, do, else, end, mod, or, then, to }
AddOp	{ +, -, or }	{ (, false, id, not, num, string, true }
ArrayType	{ array }	{ ; }
AssignOrCall	{ id }	{ ;, else, end }
AssignOrCallTail	{ (, :=, [, epsilon }	{ ;, else, end }
CompoundStmt	{ begin }	{ ., ;, else, end }
ConstDecl	{ const, epsilon }	{ begin, function, procedure, var }
ConstList	{ id }	{ begin, function, procedure, var }
ConstListTail	{ epsilon, id }	{ begin, function, procedure, var }
ConstVal	{ false, num, string, true }	{ ; }
DeclPart	{ const, epsilon, function, procedure, var }	{ begin }
ElsePart	{ else, epsilon }	{ ;, else, end }
Expr	{ (, false, id, not, num, string, true }	{), ,, ;,], do, else, end, then, to }
ExprList	{ (, false, id, not, num, string, true }	{) }
ExprListTail	{ ,, epsilon }	{) }
Factor	{ (, false, id, not, num, string, true }	{), *, +, ,, -, /, :, <, ≤, ◇, =, >, ≥,], and, div, do, else, end, mod, or, then, to }
FactorIdentTail	{ (, [, epsilon }	{), *, +, ,, -, /, :, <, ≤, ◇, =, >, ≥,], and, div, do, else, end, mod, or, then, to }
FactorTail	{ *, /, and, div, epsilon, mod }	{), +, ,, -, ;, <, ≤, ◇, =, >, ≥,], do, else, end, or, then, to }
ForStmt	{ for }	{ ;, else, end }
FormalList	{ id }	{) }
FormalListTail	{ ;, epsilon }	{) }
FormalParams	{ (, epsilon }	{ :, ; }
FunctionHeader	{ function }	{ ; }

IdList	{ id }	{ }, : }
IdListTail	{ ,, epsilon }	{ }, : }
IfStmt	{ if }	{ ;, else, end }
MulOp	{ *, /, and, div, mod }	{ (, false, id, not, num, string, true }
ProcDecl	{ epsilon, function, procedure }	{ begin }
ProcDeclProj	{ function, procedure }	{ begin, function, procedure }
ProcedureHeader	{ procedure }	{ ; }
Program	{ program }	{ \$ }
ReadStmt	{ read, readln }	{ ;, else, end }
RelOp	{ <, ≤, >, =, >, ≥ }	{ (, false, id, not, num, string, true }
RelOpExpr	{ <, ≤, >, =, >, ≥, epsilon }	{), ,, ;,], do, else, end, then, to }
SimpleExpr	{ (, false, id, not, num, string, true }	{), ,, ;, <, ≤, >, =, >, ≥,], do, else, end, then, to }
SimpleStmt	{ epsilon, id, read, readln, write, writeln }	{ ;, else, end }
SimpleType	{ boolean, char, integer, real }	{), ; }
Stmt	{ begin, epsilon, for, id, if, read, readln, while, write, writeln }	{ ;, else, end }
StmtList	{ ;, begin, epsilon, for, id, if, read, readln, while, write, writeln }	{ end }
StmtListTail	{ ;, epsilon }	{ end }
StructuredStmt	{ begin, for, if, while }	{ ;, else, end }
Term	{ (, false, id, not, num, string, true }	{), +, ,, -, ;, <, ≤, >, =, >, ≥,], do, else, end, or, then, to }
TermTail	{ +, -, epsilon, or }	{), ,, ;, <, ≤, >, =, >, ≥,], do, else, end, then, to }
Type	{ array, boolean, char, integer, real }	{ ; }
VarDecl	{ epsilon, var }	{ begin, function, procedure }
VarList	{ id }	{ begin, function, procedure }
VarListTail	{ epsilon, id }	{ begin, function, procedure }
WhileStmt	{ while }	{ ;, else, end }
WriteStmt	{ write, writeln }	{ ;, else, end }

2. LL(1) Predictive Parsing Table

LL(1) Parsing Table Entries

Non-Terminal	Terminal Lookahead	Applied Production Rule
ActualParams	(	ActualParams $\rightarrow$ (ExprList) (Rule 52)
ActualParams	)	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	*	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	+	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	,	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	-	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	/	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	;	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	<	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	$\leq$	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	$\diamond$	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	=	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	>	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	$\geq$	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	]	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	and	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	div	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	do	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	else	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	end	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	mod	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	or	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	then	ActualParams $\rightarrow \epsilon$ (Rule 53)
ActualParams	to	ActualParams $\rightarrow \epsilon$ (Rule 53)
AddOp	+	AddOp $\rightarrow$ + (Rule 82)
AddOp	-	AddOp $\rightarrow$ - (Rule 83)

AddOp	or	AddOp $\rightarrow$ or (Rule 84)
ArrayType	array	ArrayType $\rightarrow$ array [num .. num] of SimpleType (Rule 26)
AssignOrCall	id	AssignOrCall $\rightarrow$ id AssignOrCallTail (Rule 48)
AssignOrCallTail	(	AssignOrCallTail $\rightarrow$ ActualParams (Rule 51)
AssignOrCallTail	:=	AssignOrCallTail $\rightarrow$:= Expr (Rule 49)
AssignOrCallTail	;	AssignOrCallTail $\rightarrow$ ActualParams (Rule 51)
AssignOrCallTail	[	AssignOrCallTail $\rightarrow$ [Expr] := Expr (Rule 50)
AssignOrCallTail	else	AssignOrCallTail $\rightarrow$ ActualParams (Rule 51)
AssignOrCallTail	end	AssignOrCallTail $\rightarrow$ ActualParams (Rule 51)
CompoundStmt	begin	CompoundStmt $\rightarrow$ begin StmtList end (Rule 38)
ConstDecl	begin	ConstDecl $\rightarrow$ ϵ (Rule 4)
ConstDecl	const	ConstDecl $\rightarrow$ const ConstList (Rule 3)
ConstDecl	function	ConstDecl $\rightarrow$ ϵ (Rule 4)
ConstDecl	procedure	ConstDecl $\rightarrow$ ϵ (Rule 4)
ConstDecl	var	ConstDecl $\rightarrow$ ϵ (Rule 4)
ConstList	id	ConstList $\rightarrow$ id = ConstVal ; ConstListTail (Rule 5)
ConstListTail	begin	ConstListTail $\rightarrow$ ϵ (Rule 7)
ConstListTail	function	ConstListTail $\rightarrow$ ϵ (Rule 7)
ConstListTail	id	ConstListTail $\rightarrow$ id = ConstVal ; ConstListTail (Rule 6)
ConstListTail	procedure	ConstListTail $\rightarrow$ ϵ (Rule 7)
ConstListTail	var	ConstListTail $\rightarrow$ ϵ (Rule 7)
ConstVal	false	ConstVal $\rightarrow$ false (Rule 11)
ConstVal	num	ConstVal $\rightarrow$ num (Rule 8)
ConstVal	string	ConstVal $\rightarrow$ string (Rule 9)
ConstVal	true	ConstVal $\rightarrow$ true (Rule 10)
DeclPart	begin	DeclPart $\rightarrow$ ConstDecl VarDecl ProcDecl (Rule 2)
DeclPart	const	DeclPart $\rightarrow$ ConstDecl VarDecl ProcDecl (Rule 2)
DeclPart	function	DeclPart $\rightarrow$ ConstDecl VarDecl ProcDecl (Rule 2)
DeclPart	procedure	DeclPart $\rightarrow$ ConstDecl VarDecl ProcDecl (Rule 2)
DeclPart	var	DeclPart $\rightarrow$ ConstDecl VarDecl ProcDecl (Rule 2)

ElsePart	;	ElsePart $\rightarrow \epsilon$ (Rule 67)
ElsePart	else	ElsePart $\rightarrow$ else Stmt (Rule 66)
ElsePart	end	ElsePart $\rightarrow \epsilon$ (Rule 67)
Expr	(	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
Expr	false	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
Expr	id	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
Expr	not	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
Expr	num	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
Expr	string	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
Expr	true	Expr $\rightarrow$ SimpleExpr RelOpExpr (Rule 70)
ExprList	(	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprList	false	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprList	id	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprList	not	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprList	num	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprList	string	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprList	true	ExprList $\rightarrow$ Expr ExprListTail (Rule 54)
ExprListTail	)	ExprListTail $\rightarrow \epsilon$ (Rule 56)
ExprListTail	,	ExprListTail $\rightarrow$, Expr ExprListTail (Rule 55)
Factor	(	Factor $\rightarrow$ (Expr) (Rule 98)
Factor	false	Factor $\rightarrow$ false (Rule 97)
Factor	id	Factor $\rightarrow$ id FactorIdentTail (Rule 93)
Factor	not	Factor $\rightarrow$ not Factor (Rule 99)
Factor	num	Factor $\rightarrow$ num (Rule 94)
Factor	string	Factor $\rightarrow$ string (Rule 95)
Factor	true	Factor $\rightarrow$ true (Rule 96)
FactorIdentTail	(	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	)	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	*	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	+	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)

FactorIdentTail	,	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	-	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	/	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	;	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	<	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	$\leq$	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	$\langle \rangle$	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	=	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	>	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	$\geq$	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	[	FactorIdentTail $\rightarrow$ [Expr] (Rule 100)
FactorIdentTail	]	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	and	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	div	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	do	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	else	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	end	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	mod	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	or	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	then	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorIdentTail	to	FactorIdentTail $\rightarrow$ ActualParams (Rule 101)
FactorTail	)	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	*	FactorTail $\rightarrow$ MulOp Factor FactorTail (Rule 86)
FactorTail	+	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	,	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	-	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	/	FactorTail $\rightarrow$ MulOp Factor FactorTail (Rule 86)
FactorTail	;	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	<	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	$\leq$	FactorTail $\rightarrow \epsilon$ (Rule 87)

FactorTail	<>	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	=	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	>	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	$\geq$	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	]	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	and	FactorTail $\rightarrow \text{MulOp Factor FactorTail}$ (Rule 86)
FactorTail	div	FactorTail $\rightarrow \text{MulOp Factor FactorTail}$ (Rule 86)
FactorTail	do	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	else	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	end	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	mod	FactorTail $\rightarrow \text{MulOp Factor FactorTail}$ (Rule 86)
FactorTail	or	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	then	FactorTail $\rightarrow \epsilon$ (Rule 87)
FactorTail	to	FactorTail $\rightarrow \epsilon$ (Rule 87)
ForStmt	for	ForStmt $\rightarrow \text{for id := Expr to Expr do Stmt}$ (Rule 69)
FormalList	id	FormalList $\rightarrow \text{IdList : SimpleType FormalListTail}$ (Rule 35)
FormalListTail	)	FormalListTail $\rightarrow \epsilon$ (Rule 37)
FormalListTail	;	FormalListTail $\rightarrow \text{; IdList : SimpleType FormalListTail}$ (Rule 36)
FormalParams	(	FormalParams $\rightarrow (\text{FormalList})$ (Rule 33)
FormalParams	:	FormalParams $\rightarrow \epsilon$ (Rule 34)
FormalParams	;	FormalParams $\rightarrow \epsilon$ (Rule 34)
FunctionHeader	function	FunctionHeader $\rightarrow \text{function id FormalParams : SimpleType}$ (Rule 32)
IdList	id	IdList $\rightarrow \text{id IdListTail}$ (Rule 17)
IdListTail	)	IdListTail $\rightarrow \epsilon$ (Rule 19)
IdListTail	,	IdListTail $\rightarrow \text{, id IdListTail}$ (Rule 18)
IdListTail	:	IdListTail $\rightarrow \epsilon$ (Rule 19)
IfStmt	if	IfStmt $\rightarrow \text{if Expr then Stmt ElsePart}$ (Rule 65)
MulOp	*	MulOp $\rightarrow *$ (Rule 88)
MulOp	/	MulOp $\rightarrow /$ (Rule 89)

MulOp	and	MulOp $\rightarrow$ and (Rule 92)
MulOp	div	MulOp $\rightarrow$ div (Rule 90)
MulOp	mod	MulOp $\rightarrow$ mod (Rule 91)
ProcDecl	begin	ProcDecl $\rightarrow \epsilon$ (Rule 28)
ProcDecl	function	ProcDecl $\rightarrow$ ProcDeclProj ProcDecl (Rule 27)
ProcDecl	procedure	ProcDecl $\rightarrow$ ProcDeclProj ProcDecl (Rule 27)
ProcDeclProj	function	ProcDeclProj $\rightarrow$ FunctionHeader ; DeclPart CompoundStmt ; (Rule 30)
ProcDeclProj	procedure	ProcDeclProj $\rightarrow$ ProcedureHeader ; DeclPart CompoundStmt ; (Rule 29)
ProcedureHeader	procedure	ProcedureHeader $\rightarrow$ procedure id FormalParams (Rule 31)
Program	program	Program $\rightarrow$ program id ; DeclPart CompoundStmt . (Rule 1)
ReadStmt	read	ReadStmt $\rightarrow$ read (IdList) (Rule 60)
ReadStmt	readLn	ReadStmt $\rightarrow$ readLn (IdList) (Rule 59)
RelOp	<	RelOp $\rightarrow$ < (Rule 75)
RelOp	$\leq$	RelOp $\rightarrow$ <= (Rule 76)
RelOp	$\diamond$	RelOp $\rightarrow$ <> (Rule 74)
RelOp	=	RelOp $\rightarrow$ = (Rule 73)
RelOp	>	RelOp $\rightarrow$ > (Rule 77)
RelOp	$\geq$	RelOp $\rightarrow$ >= (Rule 78)
RelOpExpr	)	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	,	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	;	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	<	RelOpExpr $\rightarrow$ RelOp SimpleExpr (Rule 71)
RelOpExpr	$\leq$	RelOpExpr $\rightarrow$ RelOp SimpleExpr (Rule 71)
RelOpExpr	$\diamond$	RelOpExpr $\rightarrow$ RelOp SimpleExpr (Rule 71)
RelOpExpr	=	RelOpExpr $\rightarrow$ RelOp SimpleExpr (Rule 71)
RelOpExpr	>	RelOpExpr $\rightarrow$ RelOp SimpleExpr (Rule 71)
RelOpExpr	$\geq$	RelOpExpr $\rightarrow$ RelOp SimpleExpr (Rule 71)
RelOpExpr	]	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	do	RelOpExpr $\rightarrow \epsilon$ (Rule 72)

RelOpExpr	else	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	end	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	then	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
RelOpExpr	to	RelOpExpr $\rightarrow \epsilon$ (Rule 72)
SimpleExpr	(	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleExpr	false	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleExpr	id	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleExpr	not	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleExpr	num	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleExpr	string	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleExpr	true	SimpleExpr $\rightarrow$ Term TermTail (Rule 79)
SimpleStmt	;	SimpleStmt $\rightarrow \epsilon$ (Rule 47)
SimpleStmt	else	SimpleStmt $\rightarrow \epsilon$ (Rule 47)
SimpleStmt	end	SimpleStmt $\rightarrow \epsilon$ (Rule 47)
SimpleStmt	id	SimpleStmt $\rightarrow$ AssignOrCall (Rule 44)
SimpleStmt	read	SimpleStmt $\rightarrow$ ReadStmt (Rule 46)
SimpleStmt	readLn	SimpleStmt $\rightarrow$ ReadStmt (Rule 46)
SimpleStmt	write	SimpleStmt $\rightarrow$ WriteStmt (Rule 45)
SimpleStmt	writeln	SimpleStmt $\rightarrow$ WriteStmt (Rule 45)
SimpleType	boolean	SimpleType $\rightarrow$ boolean (Rule 24)
SimpleType	char	SimpleType $\rightarrow$ char (Rule 25)
SimpleType	integer	SimpleType $\rightarrow$ integer (Rule 22)
SimpleType	real	SimpleType $\rightarrow$ real (Rule 23)
Stmt	;	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	begin	Stmt $\rightarrow$ StructuredStmt (Rule 43)
Stmt	else	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	end	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	for	Stmt $\rightarrow$ StructuredStmt (Rule 43)
Stmt	id	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	if	Stmt $\rightarrow$ StructuredStmt (Rule 43)

Stmt	read	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	readLn	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	while	Stmt $\rightarrow$ StructuredStmt (Rule 43)
Stmt	write	Stmt $\rightarrow$ SimpleStmt (Rule 42)
Stmt	writeln	Stmt $\rightarrow$ SimpleStmt (Rule 42)
StmtList	;	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	begin	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	end	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	for	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	id	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	if	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	read	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	readLn	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	while	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	write	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtList	writeln	StmtList $\rightarrow$ Stmt StmtListTail (Rule 39)
StmtListTail	;	StmtListTail $\rightarrow$; Stmt StmtListTail (Rule 40)
StmtListTail	end	StmtListTail $\rightarrow$ ϵ (Rule 41)
StructuredStmt	begin	StructuredStmt $\rightarrow$ CompoundStmt (Rule 61)
StructuredStmt	for	StructuredStmt $\rightarrow$ ForStmt (Rule 64)
StructuredStmt	if	StructuredStmt $\rightarrow$ IfStmt (Rule 62)
StructuredStmt	while	StructuredStmt $\rightarrow$ WhileStmt (Rule 63)
Term	(	Term $\rightarrow$ Factor FactorTail (Rule 85)
Term	false	Term $\rightarrow$ Factor FactorTail (Rule 85)
Term	id	Term $\rightarrow$ Factor FactorTail (Rule 85)
Term	not	Term $\rightarrow$ Factor FactorTail (Rule 85)
Term	num	Term $\rightarrow$ Factor FactorTail (Rule 85)
Term	string	Term $\rightarrow$ Factor FactorTail (Rule 85)
Term	true	Term $\rightarrow$ Factor FactorTail (Rule 85)
TermTail	)	TermTail $\rightarrow$ ϵ (Rule 81)

TermTail	+	TermTail $\rightarrow$ AddOp Term TermTail (Rule 80)
TermTail	,	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	-	TermTail $\rightarrow$ AddOp Term TermTail (Rule 80)
TermTail	;	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	<	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	$\leq$	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	$\lt$	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	=	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	>	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	$\geq$	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	]	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	do	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	else	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	end	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	or	TermTail $\rightarrow$ AddOp Term TermTail (Rule 80)
TermTail	then	TermTail $\rightarrow \epsilon$ (Rule 81)
TermTail	to	TermTail $\rightarrow \epsilon$ (Rule 81)
Type	array	Type $\rightarrow$ ArrayType (Rule 21)
Type	boolean	Type $\rightarrow$ SimpleType (Rule 20)
Type	char	Type $\rightarrow$ SimpleType (Rule 20)
Type	integer	Type $\rightarrow$ SimpleType (Rule 20)
Type	real	Type $\rightarrow$ SimpleType (Rule 20)
VarDecl	begin	VarDecl $\rightarrow \epsilon$ (Rule 13)
VarDecl	function	VarDecl $\rightarrow \epsilon$ (Rule 13)
VarDecl	procedure	VarDecl $\rightarrow \epsilon$ (Rule 13)
VarDecl	var	VarDecl $\rightarrow$ var VarList (Rule 12)
VarList	id	VarList $\rightarrow$ IdList : Type ; VarListTail (Rule 14)
VarListTail	begin	VarListTail $\rightarrow \epsilon$ (Rule 16)
VarListTail	function	VarListTail $\rightarrow \epsilon$ (Rule 16)
VarListTail	id	VarListTail $\rightarrow$ IdList : Type ; VarListTail (Rule 15)

VarListTail	procedure	VarListTail $\rightarrow \epsilon$ (Rule 16)
WhileStmt	while	WhileStmt $\rightarrow$ while Expr do Stmt (Rule 68)
WriteStmt	write	WriteStmt $\rightarrow$ write (ExprList) (Rule 58)
WriteStmt	writeln	WriteStmt $\rightarrow$ writeln (ExprList) (Rule 57)

3. SLR(1) Parsing Table

SLR(1) Action & Goto Table (Sparse Representation)

State	Lookahead Symbol	Action / Goto Decision
State 0	program	Shift to State 2
State 0	Program	Goto State 1
State 0	Program	Goto State 1
State 1	\$	ACCEPT
State 2	id	Shift to State 3
State 3	;	Shift to State 4
State 4	begin	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 4	const	Shift to State 7
State 4	function	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 4	procedure	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 4	var	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 4	ConstDecl	Goto State 5
State 4	DeclPart	Goto State 6
State 5	begin	Reduce by Rule 13 (VarDecl $\rightarrow \epsilon$)
State 5	function	Reduce by Rule 13 (VarDecl $\rightarrow \epsilon$)
State 5	procedure	Reduce by Rule 13 (VarDecl $\rightarrow \epsilon$)
State 5	var	Shift to State 9
State 5	VarDecl	Goto State 8
State 6	begin	Shift to State 11
State 6	CompoundStmt	Goto State 10
State 7	id	Shift to State 13

State 7	ConstList	Goto State 12
State 8	begin	Reduce by Rule 28 (ProcDecl $\rightarrow \epsilon$)
State 8	function	Shift to State 18
State 8	procedure	Shift to State 19
State 8	FunctionHeader	Goto State 14
State 8	ProcDecl	Goto State 15
State 8	ProcDeclProj	Goto State 16
State 8	ProcedureHeader	Goto State 17
State 9	id	Shift to State 22
State 9	IdList	Goto State 20
State 9	VarList	Goto State 21
State 10	.	Shift to State 23
State 11	;	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 11	begin	Shift to State 11
State 11	else	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 11	end	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 11	for	Shift to State 35
State 11	id	Shift to State 36
State 11	if	Shift to State 37
State 11	read	Shift to State 38
State 11	readln	Shift to State 39
State 11	while	Shift to State 40
State 11	write	Shift to State 41
State 11	writeln	Shift to State 42
State 11	AssignOrCall	Goto State 24
State 11	CompoundStmt	Goto State 25
State 11	ForStmt	Goto State 26
State 11	IfStmt	Goto State 27
State 11	ReadStmt	Goto State 28
State 11	SimpleStmt	Goto State 29

State 11	Stmt	Goto State 30
State 11	StmtList	Goto State 31
State 11	StructuredStmt	Goto State 32
State 11	WhileStmt	Goto State 33
State 11	WriteStmt	Goto State 34
State 12	begin	Reduce by Rule 3 (ConstDecl $\rightarrow$ const ConstList)
State 12	function	Reduce by Rule 3 (ConstDecl $\rightarrow$ const ConstList)
State 12	procedure	Reduce by Rule 3 (ConstDecl $\rightarrow$ const ConstList)
State 12	var	Reduce by Rule 3 (ConstDecl $\rightarrow$ const ConstList)
State 13	=	Shift to State 43
State 14	;	Shift to State 44
State 15	begin	Reduce by Rule 2 (DeclPart $\rightarrow$ ConstDecl VarDecl ProcDecl)
State 16	begin	Reduce by Rule 28 (ProcDecl $\rightarrow \epsilon$)
State 16	function	Shift to State 18
State 16	procedure	Shift to State 19
State 16	FunctionHeader	Goto State 14
State 16	ProcDecl	Goto State 45
State 16	ProcDeclProj	Goto State 16
State 16	ProcedureHeader	Goto State 17
State 17	;	Shift to State 46
State 18	id	Shift to State 47
State 19	id	Shift to State 48
State 20	:	Shift to State 49
State 21	begin	Reduce by Rule 12 (VarDecl $\rightarrow$ var VarList)
State 21	function	Reduce by Rule 12 (VarDecl $\rightarrow$ var VarList)
State 21	procedure	Reduce by Rule 12 (VarDecl $\rightarrow$ var VarList)
State 22	)	Reduce by Rule 19 (IdListTail $\rightarrow \epsilon$)
State 22	,	Shift to State 51
State 22	:	Reduce by Rule 19 (IdListTail $\rightarrow \epsilon$)
State 22	IdListTail	Goto State 50

State 23	\$	Reduce by Rule 1 (Program $\rightarrow$ program id ; DeclPart CompoundStmt .)
State 24	;	Reduce by Rule 44 (SimpleStmt $\rightarrow$ AssignOrCall)
State 24	else	Reduce by Rule 44 (SimpleStmt $\rightarrow$ AssignOrCall)
State 24	end	Reduce by Rule 44 (SimpleStmt $\rightarrow$ AssignOrCall)
State 25	;	Reduce by Rule 61 (StructuredStmt $\rightarrow$ CompoundStmt)
State 25	else	Reduce by Rule 61 (StructuredStmt $\rightarrow$ CompoundStmt)
State 25	end	Reduce by Rule 61 (StructuredStmt $\rightarrow$ CompoundStmt)
State 26	;	Reduce by Rule 64 (StructuredStmt $\rightarrow$ ForStmt)
State 26	else	Reduce by Rule 64 (StructuredStmt $\rightarrow$ ForStmt)
State 26	end	Reduce by Rule 64 (StructuredStmt $\rightarrow$ ForStmt)
State 27	;	Reduce by Rule 62 (StructuredStmt $\rightarrow$ IfStmt)
State 27	else	Reduce by Rule 62 (StructuredStmt $\rightarrow$ IfStmt)
State 27	end	Reduce by Rule 62 (StructuredStmt $\rightarrow$ IfStmt)
State 28	;	Reduce by Rule 46 (SimpleStmt $\rightarrow$ ReadStmt)
State 28	else	Reduce by Rule 46 (SimpleStmt $\rightarrow$ ReadStmt)
State 28	end	Reduce by Rule 46 (SimpleStmt $\rightarrow$ ReadStmt)
State 29	;	Reduce by Rule 42 (Stmt $\rightarrow$ SimpleStmt)
State 29	else	Reduce by Rule 42 (Stmt $\rightarrow$ SimpleStmt)
State 29	end	Reduce by Rule 42 (Stmt $\rightarrow$ SimpleStmt)
State 30	;	Shift to State 53
State 30	end	Reduce by Rule 41 (StmtListTail $\rightarrow \epsilon$)
State 30	StmtListTail	Goto State 52
State 31	end	Shift to State 54
State 32	;	Reduce by Rule 43 (Stmt $\rightarrow$ StructuredStmt)
State 32	else	Reduce by Rule 43 (Stmt $\rightarrow$ StructuredStmt)
State 32	end	Reduce by Rule 43 (Stmt $\rightarrow$ StructuredStmt)
State 33	;	Reduce by Rule 63 (StructuredStmt $\rightarrow$ WhileStmt)
State 33	else	Reduce by Rule 63 (StructuredStmt $\rightarrow$ WhileStmt)
State 33	end	Reduce by Rule 63 (StructuredStmt $\rightarrow$ WhileStmt)

State 34	;	Reduce by Rule 45 (SimpleStmt $\rightarrow$ WriteStmt)
State 34	else	Reduce by Rule 45 (SimpleStmt $\rightarrow$ WriteStmt)
State 34	end	Reduce by Rule 45 (SimpleStmt $\rightarrow$ WriteStmt)
State 35	id	Shift to State 55
State 36	(	Shift to State 58
State 36	)	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	*	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	+	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	,	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	-	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	/	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	::=	Shift to State 59
State 36	;	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	<	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	$\leq$	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	$\lt$	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	=	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	>	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	$\geq$	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	[	Shift to State 60
State 36	]	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	and	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	div	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	do	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	else	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	end	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	mod	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	or	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	then	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 36	to	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)

State 36	ActualParams	Goto State 56
State 36	AssignOrCallTail	Goto State 57
State 37	(	Shift to State 65
State 37	false	Shift to State 66
State 37	id	Shift to State 67
State 37	not	Shift to State 68
State 37	num	Shift to State 69
State 37	string	Shift to State 70
State 37	true	Shift to State 71
State 37	Expr	Goto State 61
State 37	Factor	Goto State 62
State 37	SimpleExpr	Goto State 63
State 37	Term	Goto State 64
State 38	(	Shift to State 72
State 39	(	Shift to State 73
State 40	(	Shift to State 65
State 40	false	Shift to State 66
State 40	id	Shift to State 67
State 40	not	Shift to State 68
State 40	num	Shift to State 69
State 40	string	Shift to State 70
State 40	true	Shift to State 71
State 40	Expr	Goto State 74
State 40	Factor	Goto State 62
State 40	SimpleExpr	Goto State 63
State 40	Term	Goto State 64
State 41	(	Shift to State 75
State 42	(	Shift to State 76
State 43	false	Shift to State 78
State 43	num	Shift to State 79

State 43	string	Shift to State 80
State 43	true	Shift to State 81
State 43	ConstVal	Goto State 77
State 44	begin	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 44	const	Shift to State 7
State 44	function	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 44	procedure	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 44	var	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 44	ConstDecl	Goto State 5
State 44	DeclPart	Goto State 82
State 45	begin	Reduce by Rule 27 (ProcDecl $\rightarrow$ ProcDeclProj ProcDecl)
State 46	begin	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 46	const	Shift to State 7
State 46	function	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 46	procedure	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 46	var	Reduce by Rule 4 (ConstDecl $\rightarrow \epsilon$)
State 46	ConstDecl	Goto State 5
State 46	DeclPart	Goto State 83
State 47	(	Shift to State 85
State 47	:	Reduce by Rule 34 (FormalParams $\rightarrow \epsilon$)
State 47	;	Reduce by Rule 34 (FormalParams $\rightarrow \epsilon$)
State 47	FormalParams	Goto State 84
State 48	(	Shift to State 85
State 48	:	Reduce by Rule 34 (FormalParams $\rightarrow \epsilon$)
State 48	;	Reduce by Rule 34 (FormalParams $\rightarrow \epsilon$)
State 48	FormalParams	Goto State 86
State 49	array	Shift to State 90
State 49	boolean	Shift to State 91
State 49	char	Shift to State 92
State 49	integer	Shift to State 93

State 49	real	Shift to State 94
State 49	ArrayType	Goto State 87
State 49	SimpleType	Goto State 88
State 49	Type	Goto State 89
State 50	)	Reduce by Rule 17 (IdList $\rightarrow$ id IdListTail)
State 50	:	Reduce by Rule 17 (IdList $\rightarrow$ id IdListTail)
State 51	id	Shift to State 95
State 52	end	Reduce by Rule 39 (StmtList $\rightarrow$ Stmt StmtListTail)
State 53	;	Reduce by Rule 47 (SimpleStmt $\rightarrow$ ϵ)
State 53	begin	Shift to State 11
State 53	else	Reduce by Rule 47 (SimpleStmt $\rightarrow$ ϵ)
State 53	end	Reduce by Rule 47 (SimpleStmt $\rightarrow$ ϵ)
State 53	for	Shift to State 35
State 53	id	Shift to State 36
State 53	if	Shift to State 37
State 53	read	Shift to State 38
State 53	readln	Shift to State 39
State 53	while	Shift to State 40
State 53	write	Shift to State 41
State 53	writeln	Shift to State 42
State 53	AssignOrCall	Goto State 24
State 53	CompoundStmt	Goto State 25
State 53	ForStmt	Goto State 26
State 53	IfStmt	Goto State 27
State 53	ReadStmt	Goto State 28
State 53	SimpleStmt	Goto State 29
State 53	Stmt	Goto State 96
State 53	StructuredStmt	Goto State 32
State 53	WhileStmt	Goto State 33
State 53	WriteStmt	Goto State 34

State 54	.	Reduce by Rule 38 (CompoundStmt → begin StmtList end)
State 54	;	Reduce by Rule 38 (CompoundStmt → begin StmtList end)
State 54	else	Reduce by Rule 38 (CompoundStmt → begin StmtList end)
State 54	end	Reduce by Rule 38 (CompoundStmt → begin StmtList end)
State 55	:=	Shift to State 97
State 56	;	Reduce by Rule 51 (AssignOrCallTail → ActualParams)
State 56	else	Reduce by Rule 51 (AssignOrCallTail → ActualParams)
State 56	end	Reduce by Rule 51 (AssignOrCallTail → ActualParams)
State 57	;	Reduce by Rule 48 (AssignOrCall → id AssignOrCallTail)
State 57	else	Reduce by Rule 48 (AssignOrCall → id AssignOrCallTail)
State 57	end	Reduce by Rule 48 (AssignOrCall → id AssignOrCallTail)
State 58	(	Shift to State 65
State 58	false	Shift to State 66
State 58	id	Shift to State 67
State 58	not	Shift to State 68
State 58	num	Shift to State 69
State 58	string	Shift to State 70
State 58	true	Shift to State 71
State 58	Expr	Goto State 98
State 58	ExprList	Goto State 99
State 58	Factor	Goto State 62
State 58	SimpleExpr	Goto State 63
State 58	Term	Goto State 64
State 59	(	Shift to State 65
State 59	false	Shift to State 66
State 59	id	Shift to State 67
State 59	not	Shift to State 68
State 59	num	Shift to State 69
State 59	string	Shift to State 70
State 59	true	Shift to State 71

State 59	Expr	Goto State 100
State 59	Factor	Goto State 62
State 59	SimpleExpr	Goto State 63
State 59	Term	Goto State 64
State 60	(	Shift to State 65
State 60	false	Shift to State 66
State 60	id	Shift to State 67
State 60	not	Shift to State 68
State 60	num	Shift to State 69
State 60	string	Shift to State 70
State 60	true	Shift to State 71
State 60	Expr	Goto State 101
State 60	Factor	Goto State 62
State 60	SimpleExpr	Goto State 63
State 60	Term	Goto State 64
State 61	then	Shift to State 102
State 62	)	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	*	Shift to State 105
State 62	+	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	,	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	-	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	/	Shift to State 106
State 62	;	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	<	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	$\leq$	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	$\diamond$	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	=	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	>	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	$\geq$	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	]	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)

State 62	and	Shift to State 107
State 62	div	Shift to State 108
State 62	do	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	else	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	end	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	mod	Shift to State 109
State 62	or	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	then	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	to	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 62	FactorTail	Goto State 103
State 62	MulOp	Goto State 104
State 63	)	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	,	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	;	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	<	Shift to State 112
State 63	$\leq$	Shift to State 113
State 63	$\lt$	Shift to State 114
State 63	=	Shift to State 115
State 63	>	Shift to State 116
State 63	$\geq$	Shift to State 117
State 63	]	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	do	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	else	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	end	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	then	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	to	Reduce by Rule 72 (RelOpExpr $\rightarrow \epsilon$)
State 63	RelOp	Goto State 110
State 63	RelOpExpr	Goto State 111
State 64	)	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	+	Shift to State 120

State 64	,	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	-	Shift to State 121
State 64	;	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	<	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	$\leq$	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	$\diamond$	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	=	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	>	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	$\geq$	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	]	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	do	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	else	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	end	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	or	Shift to State 122
State 64	then	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	to	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 64	AddOp	Goto State 118
State 64	TermTail	Goto State 119
State 65	(	Shift to State 65
State 65	false	Shift to State 66
State 65	id	Shift to State 67
State 65	not	Shift to State 68
State 65	num	Shift to State 69
State 65	string	Shift to State 70
State 65	true	Shift to State 71
State 65	Expr	Goto State 123
State 65	Factor	Goto State 62
State 65	SimpleExpr	Goto State 63
State 65	Term	Goto State 64
State 66	)	Reduce by Rule 97 (Factor $\rightarrow$ false)

State 66	*	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	+	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	,	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	-	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	/	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	;	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	<	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	$\leq$	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	$\diamond$	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	=	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	>	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	$\geq$	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	]	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	and	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	div	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	do	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	else	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	end	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	mod	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	or	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	then	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 66	to	Reduce by Rule 97 (Factor $\rightarrow$ false)
State 67	(	Shift to State 58
State 67	)	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 67	*	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 67	+	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 67	,	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 67	-	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 67	/	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)
State 67	;	Reduce by Rule 53 (ActualParams $\rightarrow$ ϵ)

State 67	<	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	$\leq$	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	$\diamond$	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	=	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	>	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	$\geq$	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	[	Shift to State 126
State 67	]	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	and	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	div	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	do	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	else	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	end	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	mod	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	or	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	then	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	to	Reduce by Rule 53 (ActualParams $\rightarrow \epsilon$)
State 67	ActualParams	Goto State 124
State 67	FactorIdentTail	Goto State 125
State 68	(	Shift to State 65
State 68	false	Shift to State 66
State 68	id	Shift to State 67
State 68	not	Shift to State 68
State 68	num	Shift to State 69
State 68	string	Shift to State 70
State 68	true	Shift to State 71
State 68	Factor	Goto State 127
State 69	)	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	*	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	+	Reduce by Rule 94 (Factor $\rightarrow$ num)

State 69	,	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	-	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	/	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	;	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	<	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	$\leq$	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	$\diamond$	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	=	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	>	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	$\geq$	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	]	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	and	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	div	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	do	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	eLse	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	end	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	mod	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	or	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	then	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 69	to	Reduce by Rule 94 (Factor $\rightarrow$ num)
State 70	)	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	*	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	+	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	,	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	-	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	/	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	;	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	<	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	$\leq$	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	$\diamond$	Reduce by Rule 95 (Factor $\rightarrow$ string)

State 70	=	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	>	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	$\geq$	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	]	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	and	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	div	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	do	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	else	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	end	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	mod	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	or	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	then	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 70	to	Reduce by Rule 95 (Factor $\rightarrow$ string)
State 71	)	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	*	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	+	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	,	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	-	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	/	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	;	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	<	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	$\leq$	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	$\leq$	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	$\leq$	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	=	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	>	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	$\geq$	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	]	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	and	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	div	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	do	Reduce by Rule 96 (Factor $\rightarrow$ true)

State 71	else	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	end	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	mod	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	or	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	then	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 71	to	Reduce by Rule 96 (Factor $\rightarrow$ true)
State 72	id	Shift to State 22
State 72	IdList	Goto State 128
State 73	id	Shift to State 22
State 73	IdList	Goto State 129
State 74	do	Shift to State 130
State 75	(	Shift to State 65
State 75	false	Shift to State 66
State 75	id	Shift to State 67
State 75	not	Shift to State 68
State 75	num	Shift to State 69
State 75	string	Shift to State 70
State 75	true	Shift to State 71
State 75	Expr	Goto State 98
State 75	ExprList	Goto State 131
State 75	Factor	Goto State 62
State 75	SimpleExpr	Goto State 63
State 75	Term	Goto State 64
State 76	(	Shift to State 65
State 76	false	Shift to State 66
State 76	id	Shift to State 67
State 76	not	Shift to State 68
State 76	num	Shift to State 69
State 76	string	Shift to State 70
State 76	true	Shift to State 71

State 76	Expr	Goto State 98
State 76	ExprList	Goto State 132
State 76	Factor	Goto State 62
State 76	SimpleExpr	Goto State 63
State 76	Term	Goto State 64
State 77	;	Shift to State 133
State 78	;	Reduce by Rule 11 (ConstVal $\rightarrow$ false)
State 79	;	Reduce by Rule 8 (ConstVal $\rightarrow$ num)
State 80	;	Reduce by Rule 9 (ConstVal $\rightarrow$ string)
State 81	;	Reduce by Rule 10 (ConstVal $\rightarrow$ true)
State 82	begin	Shift to State 11
State 82	CompoundStmt	Goto State 134
State 83	begin	Shift to State 11
State 83	CompoundStmt	Goto State 135
State 84	:	Shift to State 136
State 85	id	Shift to State 22
State 85	FormalList	Goto State 137
State 85	IdList	Goto State 138
State 86	;	Reduce by Rule 31 (ProcedureHeader $\rightarrow$ procedure id FormalParams)
State 87	;	Reduce by Rule 21 (Type $\rightarrow$ ArrayType)
State 88	;	Reduce by Rule 20 (Type $\rightarrow$ SimpleType)
State 89	;	Shift to State 139
State 90	[	Shift to State 140
State 91	)	Reduce by Rule 24 (SimpleType $\rightarrow$ boolean)
State 91	;	Reduce by Rule 24 (SimpleType $\rightarrow$ boolean)
State 92	)	Reduce by Rule 25 (SimpleType $\rightarrow$ char)
State 92	;	Reduce by Rule 25 (SimpleType $\rightarrow$ char)
State 93	)	Reduce by Rule 22 (SimpleType $\rightarrow$ integer)
State 93	;	Reduce by Rule 22 (SimpleType $\rightarrow$ integer)

State 94	)	Reduce by Rule 23 ($\text{SimpleType} \rightarrow \text{real}$)
State 94	;	Reduce by Rule 23 ($\text{SimpleType} \rightarrow \text{real}$)
State 95	)	Reduce by Rule 19 ($\text{IdListTail} \rightarrow \epsilon$)
State 95	,	Shift to State 51
State 95	:	Reduce by Rule 19 ($\text{IdListTail} \rightarrow \epsilon$)
State 95	IdListTail	Goto State 141
State 96	;	Shift to State 53
State 96	end	Reduce by Rule 41 ($\text{StmntListTail} \rightarrow \epsilon$)
State 96	StmntListTail	Goto State 142
State 97	(	Shift to State 65
State 97	false	Shift to State 66
State 97	id	Shift to State 67
State 97	not	Shift to State 68
State 97	num	Shift to State 69
State 97	string	Shift to State 70
State 97	true	Shift to State 71
State 97	Expr	Goto State 143
State 97	Factor	Goto State 62
State 97	SimpleExpr	Goto State 63
State 97	Term	Goto State 64
State 98	)	Reduce by Rule 56 ($\text{ExprListTail} \rightarrow \epsilon$)
State 98	,	Shift to State 145
State 98	ExprListTail	Goto State 144
State 99	)	Shift to State 146
State 100	;	Reduce by Rule 49 ($\text{AssignOrCallTail} \rightarrow := \text{Expr}$)
State 100	else	Reduce by Rule 49 ($\text{AssignOrCallTail} \rightarrow := \text{Expr}$)
State 100	end	Reduce by Rule 49 ($\text{AssignOrCallTail} \rightarrow := \text{Expr}$)

State 101	]	Shift to State 147
State 102	;	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 102	begin	Shift to State 11
State 102	else	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 102	end	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 102	for	Shift to State 35
State 102	id	Shift to State 36
State 102	if	Shift to State 37
State 102	read	Shift to State 38
State 102	readln	Shift to State 39
State 102	while	Shift to State 40
State 102	write	Shift to State 41
State 102	writeln	Shift to State 42
State 102	AssignOrCall	Goto State 24
State 102	CompoundStmt	Goto State 25
State 102	ForStmt	Goto State 26
State 102	IfStmt	Goto State 27
State 102	ReadStmt	Goto State 28
State 102	SimpleStmt	Goto State 29

State 102	Stmt	Goto State 148
State 102	StructuredStmt	Goto State 32
State 102	WhileStmt	Goto State 33
State 102	WriteStmt	Goto State 34
State 103	)	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	+	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	,	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	-	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	;	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	<	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	$\leq$	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	$\diamond$	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	=	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	>	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	$\geq$	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	]	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	do	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	else	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	end	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)

State 103	or	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	then	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 103	to	Reduce by Rule 85 (Term $\rightarrow$ Factor FactorTail)
State 104	(	Shift to State 65
State 104	false	Shift to State 66
State 104	id	Shift to State 67
State 104	not	Shift to State 68
State 104	num	Shift to State 69
State 104	string	Shift to State 70
State 104	true	Shift to State 71
State 104	Factor	Goto State 149
State 105	(	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 105	false	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 105	id	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 105	not	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 105	num	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 105	string	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 105	true	Reduce by Rule 88 (MulOp $\rightarrow$ *)
State 106	(	Reduce by Rule 89 (MulOp $\rightarrow$ /)

State 106	false	Reduce by Rule 89 (MulOp $\rightarrow$ /)
State 106	id	Reduce by Rule 89 (MulOp $\rightarrow$ /)
State 106	not	Reduce by Rule 89 (MulOp $\rightarrow$ /)
State 106	num	Reduce by Rule 89 (MulOp $\rightarrow$ /)
State 106	string	Reduce by Rule 89 (MulOp $\rightarrow$ /)
State 106	true	Reduce by Rule 89 (MulOp $\rightarrow$ /)
State 107	(	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 107	false	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 107	id	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 107	not	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 107	num	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 107	string	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 107	true	Reduce by Rule 92 (MulOp $\rightarrow$ and)
State 108	(	Reduce by Rule 90 (MulOp $\rightarrow$ div)
State 108	false	Reduce by Rule 90 (MulOp $\rightarrow$ div)
State 108	id	Reduce by Rule 90 (MulOp $\rightarrow$ div)
State 108	not	Reduce by Rule 90 (MulOp $\rightarrow$ div)
State 108	num	Reduce by Rule 90 (MulOp $\rightarrow$ div)
State 108	string	Reduce by Rule 90 (MulOp $\rightarrow$ div)

State 108	true	Reduce by Rule 90 (MulOp $\rightarrow$ div)
State 109	(	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 109	false	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 109	id	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 109	not	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 109	num	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 109	string	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 109	true	Reduce by Rule 91 (MulOp $\rightarrow$ mod)
State 110	(	Shift to State 65
State 110	false	Shift to State 66
State 110	id	Shift to State 67
State 110	not	Shift to State 68
State 110	num	Shift to State 69
State 110	string	Shift to State 70
State 110	true	Shift to State 71
State 110	Factor	Goto State 62
State 110	SimpleExpr	Goto State 150
State 110	Term	Goto State 64
State 111	)	Reduce by Rule 70 (Expr $\rightarrow$ SimpleExpr RelOpExpr)

State 111	,	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	;	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	]	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	do	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	else	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	end	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	then	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 111	to	Reduce by Rule 70 ($\text{Expr} \rightarrow \text{SimpleExpr RelOpExpr}$)
State 112	(	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 112	false	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 112	id	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 112	not	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 112	num	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 112	string	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 112	true	Reduce by Rule 75 ($\text{RelOp} \rightarrow <$)
State 113	(	Reduce by Rule 76 ($\text{RelOp} \rightarrow \leq$)
State 113	false	Reduce by Rule 76 ($\text{RelOp} \rightarrow \leq$)
State 113	id	Reduce by Rule 76 ($\text{RelOp} \rightarrow \leq$)
State 113	not	Reduce by Rule 76 ($\text{RelOp} \rightarrow \leq$)

State 113	num	Reduce by Rule 76 (RelOp $\rightarrow$ $\leq$)
State 113	string	Reduce by Rule 76 (RelOp $\rightarrow$ $\leq$)
State 113	true	Reduce by Rule 76 (RelOp $\rightarrow$ $\leq$)
State 114	(	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 114	false	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 114	id	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 114	not	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 114	num	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 114	string	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 114	true	Reduce by Rule 74 (RelOp $\rightarrow$ $\langle \rangle$)
State 115	(	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 115	false	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 115	id	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 115	not	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 115	num	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 115	string	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 115	true	Reduce by Rule 73 (RelOp $\rightarrow$ =)
State 116	(	Reduce by Rule 77 (RelOp $\rightarrow$ >)
State 116	false	Reduce by Rule 77 (RelOp $\rightarrow$ >)

State 116	id	Reduce by Rule 77 (RelOp $\rightarrow$ >)
State 116	not	Reduce by Rule 77 (RelOp $\rightarrow$ >)
State 116	num	Reduce by Rule 77 (RelOp $\rightarrow$ >)
State 116	string	Reduce by Rule 77 (RelOp $\rightarrow$ >)
State 116	true	Reduce by Rule 77 (RelOp $\rightarrow$ >)
State 117	(	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 117	false	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 117	id	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 117	not	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 117	num	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 117	string	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 117	true	Reduce by Rule 78 (RelOp $\rightarrow$ >=)
State 118	(	Shift to State 65
State 118	false	Shift to State 66
State 118	id	Shift to State 67
State 118	not	Shift to State 68
State 118	num	Shift to State 69
State 118	string	Shift to State 70
State 118	true	Shift to State 71

State 118	Factor	Goto State 62
State 118	Term	Goto State 151
State 119	)	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	,	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	;	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	<	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	$\leq$	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	$<>$	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	=	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	>	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	$\geq$	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	]	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	do	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	else	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	end	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	then	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 119	to	Reduce by Rule 79 (SimpleExpr $\rightarrow$ Term TermTail)
State 120	(	Reduce by Rule 82 (AddOp $\rightarrow$ +)
State 120	false	Reduce by Rule 82 (AddOp $\rightarrow$ +)

State 120	id	Reduce by Rule 82 (AddOp $\rightarrow$ +)
State 120	not	Reduce by Rule 82 (AddOp $\rightarrow$ +)
State 120	num	Reduce by Rule 82 (AddOp $\rightarrow$ +)
State 120	string	Reduce by Rule 82 (AddOp $\rightarrow$ +)
State 120	true	Reduce by Rule 82 (AddOp $\rightarrow$ +)
State 121	(	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 121	false	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 121	id	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 121	not	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 121	num	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 121	string	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 121	true	Reduce by Rule 83 (AddOp $\rightarrow$ -)
State 122	(	Reduce by Rule 84 (AddOp $\rightarrow$ or)
State 122	false	Reduce by Rule 84 (AddOp $\rightarrow$ or)
State 122	id	Reduce by Rule 84 (AddOp $\rightarrow$ or)
State 122	not	Reduce by Rule 84 (AddOp $\rightarrow$ or)
State 122	num	Reduce by Rule 84 (AddOp $\rightarrow$ or)
State 122	string	Reduce by Rule 84 (AddOp $\rightarrow$ or)
State 122	true	Reduce by Rule 84 (AddOp $\rightarrow$ or)

State 123	)	Shift to State 152
State 124	)	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	*	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	+	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	,	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	-	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	/	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	;	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	<	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	$\leq$	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	$<>$	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	=	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	>	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	$\geq$	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	]	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	and	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	div	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	do	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	else	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)

State 124	end	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	mod	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	or	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	then	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 124	to	Reduce by Rule 101 (FactorIdentTail $\rightarrow$ ActualParams)
State 125	)	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	*	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	+	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	,	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	-	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	/	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	;	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	<	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	$\leq$	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	$\diamond$	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	=	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	>	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	$\geq$	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	]	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)

State 125	and	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	div	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	do	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	else	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	end	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	mod	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	or	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	then	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 125	to	Reduce by Rule 93 (Factor $\rightarrow$ id FactorIdentTail)
State 126	(	Shift to State 65
State 126	false	Shift to State 66
State 126	id	Shift to State 67
State 126	not	Shift to State 68
State 126	num	Shift to State 69
State 126	string	Shift to State 70
State 126	true	Shift to State 71
State 126	Expr	Goto State 153
State 126	Factor	Goto State 62
State 126	SimpleExpr	Goto State 63

State 126	Term	Goto State 64
State 127	)	Reduce by Rule 99 (Factor → not Factor)
State 127	*	Reduce by Rule 99 (Factor → not Factor)
State 127	+	Reduce by Rule 99 (Factor → not Factor)
State 127	,	Reduce by Rule 99 (Factor → not Factor)
State 127	-	Reduce by Rule 99 (Factor → not Factor)
State 127	/	Reduce by Rule 99 (Factor → not Factor)
State 127	;	Reduce by Rule 99 (Factor → not Factor)
State 127	<	Reduce by Rule 99 (Factor → not Factor)
State 127	≤	Reduce by Rule 99 (Factor → not Factor)
State 127	>	Reduce by Rule 99 (Factor → not Factor)
State 127	=	Reduce by Rule 99 (Factor → not Factor)
State 127	>	Reduce by Rule 99 (Factor → not Factor)
State 127	≥	Reduce by Rule 99 (Factor → not Factor)
State 127	]	Reduce by Rule 99 (Factor → not Factor)
State 127	and	Reduce by Rule 99 (Factor → not Factor)
State 127	div	Reduce by Rule 99 (Factor → not Factor)
State 127	do	Reduce by Rule 99 (Factor → not Factor)
State 127	else	Reduce by Rule 99 (Factor → not Factor)

State 127	end	Reduce by Rule 99 (Factor $\rightarrow$ not Factor)
State 127	mod	Reduce by Rule 99 (Factor $\rightarrow$ not Factor)
State 127	or	Reduce by Rule 99 (Factor $\rightarrow$ not Factor)
State 127	then	Reduce by Rule 99 (Factor $\rightarrow$ not Factor)
State 127	to	Reduce by Rule 99 (Factor $\rightarrow$ not Factor)
State 128	)	Shift to State 154
State 129	)	Shift to State 155
State 130	;	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 130	begin	Shift to State 11
State 130	else	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 130	end	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 130	for	Shift to State 35
State 130	id	Shift to State 36
State 130	if	Shift to State 37
State 130	read	Shift to State 38
State 130	readln	Shift to State 39
State 130	while	Shift to State 40
State 130	write	Shift to State 41
State 130	writeln	Shift to State 42

State 130	AssignOrCall	Goto State 24
State 130	CompoundStmt	Goto State 25
State 130	ForStmt	Goto State 26
State 130	IfStmt	Goto State 27
State 130	ReadStmt	Goto State 28
State 130	SimpleStmt	Goto State 29
State 130	Stmt	Goto State 156
State 130	StructuredStmt	Goto State 32
State 130	WhileStmt	Goto State 33
State 130	WriteStmt	Goto State 34
State 131	)	Shift to State 157
State 132	)	Shift to State 158
State 133	begin	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 133	function	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 133	id	Shift to State 160
State 133	procedure	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 133	var	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 133	ConstListTail	Goto State 159
State 134	;	Shift to State 161

State 135	;	Shift to State 162
State 136	boolean	Shift to State 91
State 136	char	Shift to State 92
State 136	integer	Shift to State 93
State 136	real	Shift to State 94
State 136	SimpleType	Goto State 163
State 137	)	Shift to State 164
State 138	:	Shift to State 165
State 139	begin	Reduce by Rule 16 ($\text{VarListTail} \rightarrow \epsilon$)
State 139	function	Reduce by Rule 16 ($\text{VarListTail} \rightarrow \epsilon$)
State 139	id	Shift to State 22
State 139	procedure	Reduce by Rule 16 ($\text{VarListTail} \rightarrow \epsilon$)
State 139	IdList	Goto State 166
State 139	VarListTail	Goto State 167
State 140	num	Shift to State 168
State 141	)	Reduce by Rule 18 ($\text{IdListTail} \rightarrow , \text{id IdListTail}$)
State 141	:	Reduce by Rule 18 ($\text{IdListTail} \rightarrow , \text{id IdListTail}$)
State 142	end	Reduce by Rule 40 ($\text{StmtListTail} \rightarrow ; \text{Stmt StmtListTail}$)
State 143	to	Shift to State 169

State 144	)	Reduce by Rule 54 ($\text{ExprList} \rightarrow \text{Expr ExprListTail}$)
State 145	(	Shift to State 65
State 145	false	Shift to State 66
State 145	id	Shift to State 67
State 145	not	Shift to State 68
State 145	num	Shift to State 69
State 145	string	Shift to State 70
State 145	true	Shift to State 71
State 145	Expr	Goto State 170
State 145	Factor	Goto State 62
State 145	SimpleExpr	Goto State 63
State 145	Term	Goto State 64
State 146	)	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)
State 146	*	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)
State 146	+	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)
State 146	,	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)
State 146	-	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)
State 146	/	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)
State 146	;	Reduce by Rule 52 ($\text{ActualParams} \rightarrow (\text{ExprList})$)

State 146	<	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	$\leq$	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	$\diamond$	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	=	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	>	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	$\geq$	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	]	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	and	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	div	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	do	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	else	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	end	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	mod	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	or	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	then	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 146	to	Reduce by Rule 52 (ActualParams $\rightarrow$ (ExprList))
State 147	:=	Shift to State 171
State 148	;	Reduce by Rule 67 (ElsePart $\rightarrow$ ϵ)
State 148	else	Shift to State 173

State 148	end	Reduce by Rule 67 (ElsePart $\rightarrow \epsilon$)
State 148	ElsePart	Goto State 172
State 149	)	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	*	Shift to State 105
State 149	+	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	,	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	-	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	/	Shift to State 106
State 149	;	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	<	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	$\leq$	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	$\diamond$	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	=	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	>	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	$\geq$	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	]	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	and	Shift to State 107
State 149	div	Shift to State 108
State 149	do	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)

State 149	else	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	end	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	mod	Shift to State 109
State 149	or	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	then	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	to	Reduce by Rule 87 (FactorTail $\rightarrow \epsilon$)
State 149	FactorTail	Goto State 174
State 149	MulOp	Goto State 104
State 150	)	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	,	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	;	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	]	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	do	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	else	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	end	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	then	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 150	to	Reduce by Rule 71 (RelOpExpr $\rightarrow$ RelOp SimpleExpr)
State 151	)	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	+	Shift to State 120

State 151	,	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	-	Shift to State 121
State 151	;	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	<	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	$\leq$	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	$\diamond$	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	=	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	>	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	$\geq$	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	]	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	do	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	else	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	end	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	or	Shift to State 122
State 151	then	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	to	Reduce by Rule 81 (TermTail $\rightarrow \epsilon$)
State 151	AddOp	Goto State 118
State 151	TermTail	Goto State 175
State 152	)	Reduce by Rule 98 (Factor $\rightarrow (\text{Expr})$)

State 152	*	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	+	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	,	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	-	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	/	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	;	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	<	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	$\leq$	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	$\lt$	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	=	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	>	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	$\geq$	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	]	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	and	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	div	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	do	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	else	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	end	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	mod	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))

State 152	or	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	then	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 152	to	Reduce by Rule 98 (Factor $\rightarrow$ (Expr))
State 153	]	Shift to State 176
State 154	;	Reduce by Rule 60 (ReadStmt $\rightarrow$ read (IdList))
State 154	else	Reduce by Rule 60 (ReadStmt $\rightarrow$ read (IdList))
State 154	end	Reduce by Rule 60 (ReadStmt $\rightarrow$ read (IdList))
State 155	;	Reduce by Rule 59 (ReadStmt $\rightarrow$ readln (IdList))
State 155	else	Reduce by Rule 59 (ReadStmt $\rightarrow$ readln (IdList))
State 155	end	Reduce by Rule 59 (ReadStmt $\rightarrow$ readln (IdList))
State 156	;	Reduce by Rule 68 (WhileStmt $\rightarrow$ while Expr do Stmt)
State 156	else	Reduce by Rule 68 (WhileStmt $\rightarrow$ while Expr do Stmt)
State 156	end	Reduce by Rule 68 (WhileStmt $\rightarrow$ while Expr do Stmt)
State 157	;	Reduce by Rule 58 (WriteStmt $\rightarrow$ write (ExprList))
State 157	else	Reduce by Rule 58 (WriteStmt $\rightarrow$ write (ExprList))
State 157	end	Reduce by Rule 58 (WriteStmt $\rightarrow$ write (ExprList))
State 158	;	Reduce by Rule 57 (WriteStmt $\rightarrow$ writeln (ExprList))
State 158	else	Reduce by Rule 57 (WriteStmt $\rightarrow$ writeln (ExprList))
State 158	end	Reduce by Rule 57 (WriteStmt $\rightarrow$ writeln (ExprList))

State 159	begin	Reduce by Rule 5 (ConstList $\rightarrow$ id = ConstVal ; ConstListTail)
State 159	function	Reduce by Rule 5 (ConstList $\rightarrow$ id = ConstVal ; ConstListTail)
State 159	procedure	Reduce by Rule 5 (ConstList $\rightarrow$ id = ConstVal ; ConstListTail)
State 159	var	Reduce by Rule 5 (ConstList $\rightarrow$ id = ConstVal ; ConstListTail)
State 160	=	Shift to State 177
State 161	begin	Reduce by Rule 30 (ProcDeclProj $\rightarrow$ FunctionHeader ; DeclPart CompoundStmt ;)
State 161	function	Reduce by Rule 30 (ProcDeclProj $\rightarrow$ FunctionHeader ; DeclPart CompoundStmt ;)
State 161	procedure	Reduce by Rule 30 (ProcDeclProj $\rightarrow$ FunctionHeader ; DeclPart CompoundStmt ;)
State 162	begin	Reduce by Rule 29 (ProcDeclProj $\rightarrow$ ProcedureHeader ; DeclPart CompoundStmt ;)
State 162	function	Reduce by Rule 29 (ProcDeclProj $\rightarrow$ ProcedureHeader ; DeclPart CompoundStmt ;)
State 162	procedure	Reduce by Rule 29 (ProcDeclProj $\rightarrow$ ProcedureHeader ; DeclPart CompoundStmt ;)
State 163	;	Reduce by Rule 32 (FunctionHeader $\rightarrow$ function id FormalParams : SimpleType)
State 164	:	Reduce by Rule 33 (FormalParams $\rightarrow$ (FormalList))
State 164	;	Reduce by Rule 33 (FormalParams $\rightarrow$ (FormalList))
State 165	boolean	Shift to State 91
State 165	char	Shift to State 92
State 165	integer	Shift to State 93
State 165	real	Shift to State 94
State 165	SimpleType	Goto State 178

State 166	:	Shift to State 179
State 167	begin	Reduce by Rule 14 (VarList $\rightarrow$ IdList : Type ; VarListTail)
State 167	function	Reduce by Rule 14 (VarList $\rightarrow$ IdList : Type ; VarListTail)
State 167	procedure	Reduce by Rule 14 (VarList $\rightarrow$ IdList : Type ; VarListTail)
State 168	..	Shift to State 180
State 169	(	Shift to State 65
State 169	false	Shift to State 66
State 169	id	Shift to State 67
State 169	not	Shift to State 68
State 169	num	Shift to State 69
State 169	string	Shift to State 70
State 169	true	Shift to State 71
State 169	Expr	Goto State 181
State 169	Factor	Goto State 62
State 169	SimpleExpr	Goto State 63
State 169	Term	Goto State 64
State 170	)	Reduce by Rule 56 (ExprListTail $\rightarrow \epsilon$)
State 170	,	Shift to State 145
State 170	ExprListTail	Goto State 182

State 171	(	Shift to State 65
State 171	false	Shift to State 66
State 171	id	Shift to State 67
State 171	not	Shift to State 68
State 171	num	Shift to State 69
State 171	string	Shift to State 70
State 171	true	Shift to State 71
State 171	Expr	Goto State 183
State 171	Factor	Goto State 62
State 171	SimpleExpr	Goto State 63
State 171	Term	Goto State 64
State 172	;	Reduce by Rule 65 (IfStmt $\rightarrow$ if Expr then Stmt ElsePart)
State 172	else	Reduce by Rule 65 (IfStmt $\rightarrow$ if Expr then Stmt ElsePart)
State 172	end	Reduce by Rule 65 (IfStmt $\rightarrow$ if Expr then Stmt ElsePart)
State 173	;	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 173	begin	Shift to State 11
State 173	else	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 173	end	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 173	for	Shift to State 35

State 173	id	Shift to State 36
State 173	if	Shift to State 37
State 173	read	Shift to State 38
State 173	readln	Shift to State 39
State 173	while	Shift to State 40
State 173	write	Shift to State 41
State 173	writeln	Shift to State 42
State 173	AssignOrCall	Goto State 24
State 173	CompoundStmt	Goto State 25
State 173	ForStmt	Goto State 26
State 173	IfStmt	Goto State 27
State 173	ReadStmt	Goto State 28
State 173	SimpleStmt	Goto State 29
State 173	Stmt	Goto State 184
State 173	StructuredStmt	Goto State 32
State 173	WhileStmt	Goto State 33
State 173	WriteStmt	Goto State 34
State 174	)	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	+	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)

State 174	,	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	-	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	;	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	<	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	$\leq$	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	$\diamond$	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	=	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	>	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	$\geq$	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	]	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	do	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	else	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	end	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	or	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	then	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 174	to	Reduce by Rule 86 (FactorTail $\rightarrow$ MulOp Factor FactorTail)
State 175	)	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	,	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	;	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)

State 175	<	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	$\leq$	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	$\diamond$	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	=	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	>	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	$\geq$	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	]	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	do	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	else	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	end	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	then	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 175	to	Reduce by Rule 80 (TermTail $\rightarrow$ AddOp Term TermTail)
State 176	)	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	*	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	+	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	,	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	-	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	/	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	;	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])

State 176	<	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	$\leq$	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	$\diamond$	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	=	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	>	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	$\geq$	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	]	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	and	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	div	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	do	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	else	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	end	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	mod	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	or	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	then	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 176	to	Reduce by Rule 100 (FactorIdentTail $\rightarrow$ [Expr])
State 177	false	Shift to State 78
State 177	num	Shift to State 79
State 177	string	Shift to State 80

State 177	true	Shift to State 81
State 177	ConstVal	Goto State 185
State 178	)	Reduce by Rule 37 (FormalListTail $\rightarrow \epsilon$)
State 178	;	Shift to State 187
State 178	FormalListTail	Goto State 186
State 179	array	Shift to State 90
State 179	boolean	Shift to State 91
State 179	char	Shift to State 92
State 179	integer	Shift to State 93
State 179	real	Shift to State 94
State 179	ArrayType	Goto State 87
State 179	SimpleType	Goto State 88
State 179	Type	Goto State 188
State 180	num	Shift to State 189
State 181	do	Shift to State 190
State 182	)	Reduce by Rule 55 (ExprListTail $\rightarrow$, Expr ExprListTail)
State 183	;	Reduce by Rule 50 (AssignOrCallTail $\rightarrow$ [Expr] := Expr)
State 183	else	Reduce by Rule 50 (AssignOrCallTail $\rightarrow$ [Expr] := Expr)
State 183	end	Reduce by Rule 50 (AssignOrCallTail $\rightarrow$ [Expr] := Expr)

State 184	;	Reduce by Rule 66 (ElsePart $\rightarrow$ else Stmt)
State 184	else	Reduce by Rule 66 (ElsePart $\rightarrow$ else Stmt)
State 184	end	Reduce by Rule 66 (ElsePart $\rightarrow$ else Stmt)
State 185	;	Shift to State 191
State 186	)	Reduce by Rule 35 (FormalList $\rightarrow$ IdList : SimpleType FormalListTail)
State 187	id	Shift to State 22
State 187	IdList	Goto State 192
State 188	;	Shift to State 193
State 189	]	Shift to State 194
State 190	;	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 190	begin	Shift to State 11
State 190	else	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 190	end	Reduce by Rule 47 (SimpleStmt $\rightarrow \epsilon$)
State 190	for	Shift to State 35
State 190	id	Shift to State 36
State 190	if	Shift to State 37
State 190	read	Shift to State 38
State 190	readLn	Shift to State 39
State 190	while	Shift to State 40

State 190	write	Shift to State 41
State 190	writeln	Shift to State 42
State 190	AssignOrCall	Goto State 24
State 190	CompoundStmt	Goto State 25
State 190	ForStmt	Goto State 26
State 190	IfStmt	Goto State 27
State 190	ReadStmt	Goto State 28
State 190	SimpleStmt	Goto State 29
State 190	Stmt	Goto State 195
State 190	StructuredStmt	Goto State 32
State 190	WhileStmt	Goto State 33
State 190	WriteStmt	Goto State 34
State 191	begin	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 191	function	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 191	id	Shift to State 160
State 191	procedure	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 191	var	Reduce by Rule 7 (ConstListTail $\rightarrow \epsilon$)
State 191	ConstListTail	Goto State 196
State 192	:	Shift to State 197

State 193	begin	Reduce by Rule 16 (VarListTail $\rightarrow \epsilon$)
State 193	function	Reduce by Rule 16 (VarListTail $\rightarrow \epsilon$)
State 193	id	Shift to State 22
State 193	procedure	Reduce by Rule 16 (VarListTail $\rightarrow \epsilon$)
State 193	IdList	Goto State 166
State 193	VarListTail	Goto State 198
State 194	of	Shift to State 199
State 195	;	Reduce by Rule 69 (ForStmt $\rightarrow$ for id := Expr to Expr do Stmt)
State 195	else	Reduce by Rule 69 (ForStmt $\rightarrow$ for id := Expr to Expr do Stmt)
State 195	end	Reduce by Rule 69 (ForStmt $\rightarrow$ for id := Expr to Expr do Stmt)
State 196	begin	Reduce by Rule 6 (ConstListTail $\rightarrow$ id = ConstVal ; ConstListTail)
State 196	function	Reduce by Rule 6 (ConstListTail $\rightarrow$ id = ConstVal ; ConstListTail)
State 196	procedure	Reduce by Rule 6 (ConstListTail $\rightarrow$ id = ConstVal ; ConstListTail)
State 196	var	Reduce by Rule 6 (ConstListTail $\rightarrow$ id = ConstVal ; ConstListTail)
State 197	boolean	Shift to State 91
State 197	char	Shift to State 92
State 197	integer	Shift to State 93
State 197	real	Shift to State 94
State 197	SimpleType	Goto State 200

State 198	begin	Reduce by Rule 15 (VarListTail $\rightarrow$ IdList : Type ; VarListTail)
State 198	function	Reduce by Rule 15 (VarListTail $\rightarrow$ IdList : Type ; VarListTail)
State 198	procedure	Reduce by Rule 15 (VarListTail $\rightarrow$ IdList : Type ; VarListTail)
State 199	boolean	Shift to State 91
State 199	char	Shift to State 92
State 199	integer	Shift to State 93
State 199	real	Shift to State 94
State 199	SimpleType	Goto State 201
State 200	)	Reduce by Rule 37 (FormalListTail $\rightarrow \epsilon$)
State 200	;	Shift to State 187
State 200	FormalListTail	Goto State 202
State 201	;	Reduce by Rule 26 (ArrayType $\rightarrow$ array [num .. num] of SimpleType)
State 202	)	Reduce by Rule 36 (FormalListTail $\rightarrow$; IdList : SimpleType FormalListTail)